Ellanti Rohith

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.05
Sri Chaitanya College	Class XII (Central Board of Secondary Education)	2024	95.20%
Jawahar Navodaya Vidyalaya	Class X (Central Board of Secondary Education)	2022	95.00%

SKILLS

Libraries and Frameworks : Numpy, Scipy, Scikit-learn, XGBoost, PyTorch, NetworkX

Programming Languages: C, C++, Python, Verilog, Assembly, SQL

Fundamentals: Data Structure and Algorithms, Database Management System, OOPs

PROJECTS

C++ Search Engine

- Built a full-text search engine from scratch in C++17 indexing documents using an inverted Index data structure with postings lists and term frequency storage
- Implemented TF-IDF (Term Frequency-Inverse Document Frequency) ranking algorithm to score and sort search results by relevance
- Designed a tokenization pipeline: lowercasing, punctuation stripping, stop word removal, and Porter stemming to normalize text before indexing

Library Management System

- Designed and implemented a normalized PostgreSQL schema (7 tables, 3NF) modeling one-to-many and many-to-many relationships for a library management system
- Enforced business logic at the database layer using a SQL view, a trigger, and a transactional stored procedure automating copy-count tracking and late-fee calculation instead of handling it in application code

Graph-Based Fraud Detection System

- Built end-to-end fraud detection pipeline using XGBoost + SMOTE on imbalanced transaction data (6% fraud rate), achieving ROC-AUC of 0.98 and 98% fraud recall with 5-fold cross-validation
- Engineered graph features (PageRank, degree centrality, clustering coefficient) from a bipartite user-merchant transaction network using NetworkX, capturing structural fraud patterns invisible to tabular models
- Designed a 2-layer PyTorch LSTM trained on sliding-window user transaction sequences to detect temporal burst patterns, combined with a 3-layer feature pipeline of tabular, graph, and rolling sequence statistics

SmartCrop-AI | AI-Powered Content-Aware Image Reframer

- Engineered an end-to-end computer vision pipeline using YOLOv8 and OpenCV to automatically detect primary image subjects and prevent focal-point cropping during multi-aspect ratio conversions.
- Implemented K-Means clustering to dynamically extract dominant border color palettes, enabling seamless adaptive canvas padding without visual distortion.
- Developed an interactive web interface with Gradio for real-time image upload, custom aspect ratio selection (16:9, 9:16, 4:5), and immediate visual reframing feedback.

Parallel Eigenvalue Computation Engine (C / Multithreading)

Engine built in C to factorize dense complex matrices and calculate eigen spectra via Householder QR Decomposition.

Accelerated $O(n^3)$ linear algebra operations using multithreading to maximize CPU core utilization.

Optimized dynamic memory and pointer structures to avoid leaks during multi-pass solver iteration.

SCHOLASTIC ACHIEVEMENTS

Awarded a merit-based grant of 1 Lakh for ranking among the top candidates nationally in the LPUNEST examination

RELEVANT COURSES

CS Courses: Intro to Programming, Data Structures and Applications

Electrical Courses: Digital Systems,Control Systems, Analog Electronics,Scientific Programming for Electrical Engineers

Signal Processing Courses: Signal Processing, Digital Signal Processing Lab

Math Courses: Probability, Random Process, Machine Learning, Complex Variables

EXTRACURRICULAR

NSS Volunteer: Coordinated community social development programs, collaborating in team-based environmental cleaning drives demonstrating strong teamwork and social responsibility.

Runner-Up, Milan 2025 Filmmaking Competition : Recognized for exceptional short film acting, direction, visual editing, and post-production execution.